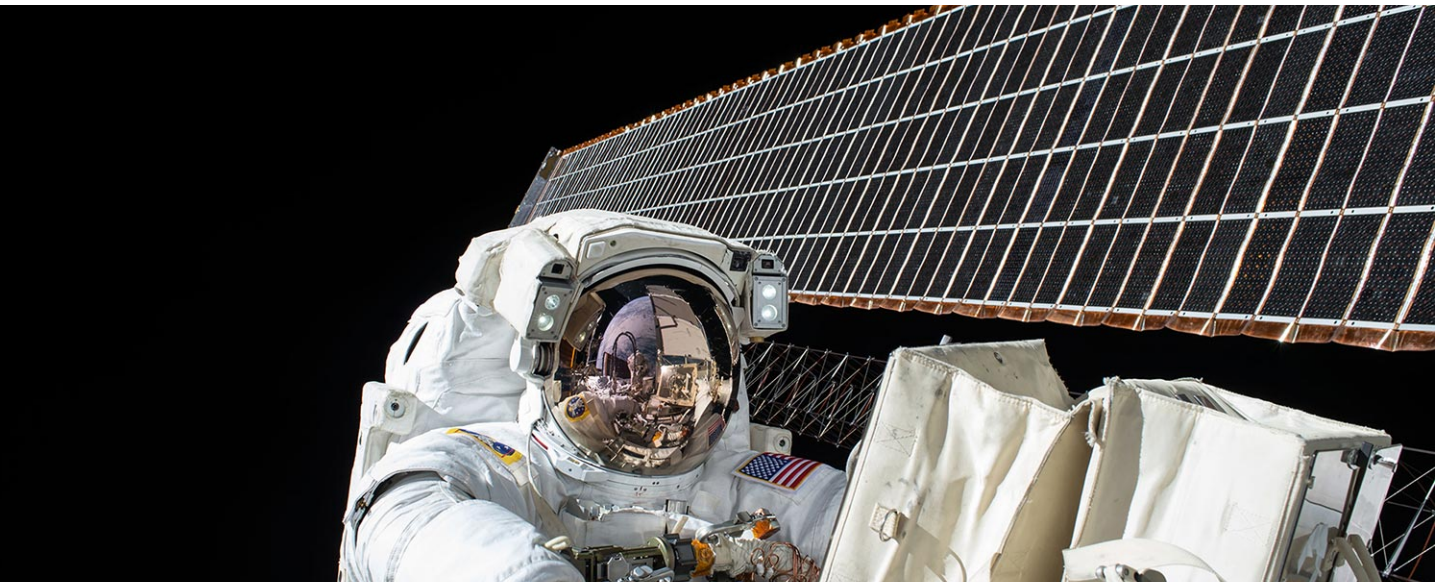


Model Question Paper -3

answer the following question 2 marks

- what is cyberethics

  Cyberethics refers to the ethical considerations and guidelines that govern behavior and interactions in the digital realm. It addresses issues related to online privacy, security, intellectual property, responsible digital citizenship, and proper use of technology.

- what is web service? give example

  A web service is a software component that provides functionalities over the internet, allowing different applications to communicate. An example is a currency conversion web service that receives currency codes and returns conversion rates.

- Mention the core technologies in web1.07

  Web 1.07 does not represent a standard version of web technologies. The concept of versions is typically applied to Web 1.0, 2.0, etc., each with its own set of technologies.

- what is front-end web development?

  Front-end web development involves designing and coding the user interface and user experience of a website. It focuses on creating elements that users interact with directly, including layouts, navigation menus, forms, and visual design.

how to create a variable in PHP? give an example

In PHP, you can create a variable by using the dollar sign ($) followed by a variable name. Variables can hold various types of data, such as numbers, strings, and arrays.

Example:

```
phpCopy code $name = "John"; $age = 25;
```

what is a column store database? give an example

A column store database is a type of database that stores data in columns rather than rows, optimizing for analytical queries and data compression. It's suitable for applications requiring efficient data aggregation and analysis.

Example: Apache Cassandra is a NoSQL database that uses a column-family data model. It is designed for managing large amounts of distributed data across multiple commodity servers.

and answer the following question with 5 marks

write the features or benefits of telnet

Features or Benefits of Telnet:
Telnet is a protocol that allows remote access to another computer over a network, typically the internet. Its features include:

- **Remote Access:** Telnet enables users to access and manage remote systems.
- **Text-Based:** It provides a command-line interface to the remote system.
- **Network Device Configuration:** Telnet is often used for configuring network devices such as routers and switches.
- **Debugging:** It helps in diagnosing network connectivity and troubleshooting issues.
- **Simplicity:** Telnet is simple to use but lacks encryption, making it less secure for sensitive data

what is HTTP cache? explain the level of cache

HTTP Cache and Levels of Cache:
HTTP caching involves storing copies of web resources to reduce server load and improve user experience. There are several levels of caching:

- **Browser Cache:** Caches resources on the user's device.
- **Proxy Cache:** Caches resources on intermediate proxy servers.
- **CDN Cache:** Content Delivery Networks cache resources on distributed servers.
- **Server Cache:** Web servers cache dynamic content for quicker retrieval.
- **Database Cache:** Database systems cache query results for faster access.

what is google pagerank? write pagerank algorithm

Google PageRank and PageRank Algorithm:
Google PageRank is an algorithm used by Google to rank web pages in search engine results. The PageRank algorithm determines a web page's importance based on the number and quality of incoming links from other pages. It follows these steps:

- Initialize each page's rank.
- Calculate a page's rank as the sum of a fraction of the rank of linking pages.
- Iterate through multiple rounds, adjusting ranks based on linking relationships.
- Convergence is reached when the ranks stabilize.

what is javascript? write the features of angularJS

JavaScript and Features of AngularJS:
JavaScript is a programming language used for adding interactivity to web pages. AngularJS is a JavaScript framework developed by Google for building dynamic web applications. AngularJS features include:

- **Data Binding:** Automatically synchronizes data between the model and view.
- **Directives:** Extend HTML with custom tags and attributes for enhanced functionality.

- **Dependency Injection:** Simplifies component management and testing.
- **MVC Architecture:** Supports the separation of concerns for easier development.
- **Templates:** Allows declarative rendering of views.

explain the model view controller architecture of angularJS

Model-View-Controller Architecture of AngularJS:
In AngularJS, the Model-View-Controller (MVC) architecture is implemented as follows:

- **Model:** Represents the application's data and logic.
- **View:** Renders the data and provides the user interface.
- **Controller:** Acts as an intermediary between the model and view, handling user interactions and updating the model.

explain text mining and sentiment analysis with an example

Text mining involves extracting valuable insights and patterns from textual data. Sentiment analysis is a subset of text mining that focuses on determining the emotional tone of a piece of text. For example, analyzing social media posts to determine whether they express positive, negative, or neutral sentiments towards a product. Machine learning techniques are often employed to classify the sentiment of text based on predefined categories.

answer the following question with 8 marks

what is cyberethics? write the important principles of cyber ethics and netiquette

1. Cyberethics, Principles, and Netiquette:

Cyberethics refers to the ethical considerations and guidelines that govern behavior and interactions in the digital realm. It addresses issues related to online privacy, security, responsible digital citizenship, intellectual property, and appropriate use of technology.

Principles of Cyberethics and Netiquette:

- **Respect for Privacy:** Respecting the privacy of individuals' personal data and information.

- **Security:** Safeguarding digital assets and data from unauthorized access and breaches.
- **Digital Responsibility:** Being accountable for one's online actions and interactions.
- **Intellectual Property:** Respecting copyright and intellectual property rights in digital content.
- **Digital Etiquette (Netiquette):**
    - Using respectful language in online communications.
    - Avoiding spamming or excessive self-promotion.
    - Properly citing and attributing sources in online content.
    - Being mindful of cultural differences in online interactions.

explain the webpage and website along with its types with suitable examples

Webpage: A webpage is a single document displayed on a web browser. It's written in HTML and may contain various types of content like text, images, videos, and interactive elements.

Website: A website is a collection of interconnected webpages under a common domain. It typically includes multiple webpages linked together through navigation menus.

Types of Websites:

- **Static Website:** Contains fixed content that rarely changes. Example: Online brochure for a company.
- **Dynamic Website:** Generates content on-the-fly based on user interactions. Example: E-commerce website showing personalized recommendations.
- **E-commerce Website:** Facilitates online buying and selling of products or services. Example: Amazon.
- **Social Media Website:** Allows users to connect, share, and interact with each other. Example: Facebook.
- **Blogging Website:** Focuses on regularly updated articles or blog posts. Example: WordPress-based blogs.

explain big data in detail. explain big data types with suitable

examples

Big Data refers to massive and complex datasets that exceed the processing capabilities of traditional database systems. It's characterized by the three Vs: Volume, Velocity, and Variety.

Types of Big Data:

- **Structured Data:** Organized data with a fixed format. Example: Sales transactions in a relational database.
- **Unstructured Data:** Data without a predefined structure. Example: Social media posts, images, and videos.
- **Semi-Structured Data:** Contains both structured and unstructured elements. Example: JSON or XML files.

explain the various elements of web development

Web development involves several elements:

- **Front-End Development:** Designing and coding the user interface using HTML, CSS, and JavaScript.
- **Back-End Development:** Implementing server-side logic and managing databases.
- **Full-Stack Development:** Combining both front-end and back-end skills.
- **Web Hosting:** Deploying the website on a web server for public access.
- **Domain Name:** The web address used to access the website.
- **Web Security:** Implementing measures to protect against cyber threats.

how to create an application in node.js? explain with an example

To create an application in Node.js:

1. Install Node.js and a text editor.
2. Create a JavaScript file (e.g., `app.js`).
3. Write your application logic using Node.js libraries.

Example - A simple HTTP server using Node.js:

```
javascriptCopy code // Import the HTTP module const http = r
equire('http'); // Create an HTTP server const server = htt
p.createServer((req, res) => { res.writeHead(200, { 'Conten
t-Type': 'text/plain' }); res.end('Hello, Node.js!'); }); //
Listen on port 3000 server.listen(3000, () => { console.log
('Server is running on port 3000'); });
```

explain the features of a non-relational database

Non-relational databases, also known as NoSQL databases, have features like:

- **Schema Flexibility:** Accommodates diverse and changing data structures without a fixed schema.
- **Horizontal Scalability:** Easily scales by distributing data across multiple servers.
- **High Performance:** Optimized for read and write operations at scale.
- **Data Variety:** Supports various data formats like documents, key-value pairs, graphs, etc.
- **No Joins:** Avoids complex joins, improving query performance.
- **Designed for Big Data:** Handles large volumes of data and high velocity.
- **Polyglot Persistence:** Supports multiple data models within a single database.